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8 February 1985
SUPERSEDING
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31 July 1978

MILITARY SPECIFICATION

SAWS, BAND, CUTOFF, HORIZONTAL, NON-ADJUSTABLE AND ADJUSTABLE CUTTING ANGLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers heavy duty, horizontal type, production metal cutting band saw machines equipped for manual stock indexing (type I) or for automatic, repetitive indexing of bar material to preset lengths (type II). Class A machines are capable of square, 90 degree cutoff. Class B machines are capable of miter sawing from 0 degree to 45 degrees cutting angle. The machines shall be suitable for cutting-off a variety of metal bars including, but not limited to, solid bars, tubing, pipe, structural shapes, and extrusions within the capacity of the machine.

1.2 Classification. This specification covers horizontal cutoff type band saws of the following types, classes, and sizes. The type, class, and size required for specific procurements shall be as specified (see 6.2.1).

Type I - Non-automatic - Manual stock feed

Class A - Non-adjustable cutting angle
(90 degrees from workpiece axis)

Size 1 - 12 x 12-inch workpiece capacity
Size 2 - 12 x 16-inch workpiece capacity
Size 3 - 12 x 20-inch workpiece capacity
Size 4 - 14 x 17-inch workpiece capacity
Size 5 - 15 x 20-inch workpiece capacity

Class B - Adjustable cutting angle
(90 degrees to 45 degrees from workpiece axis)

Size 2 - 12 x 16-inch workpiece capacity

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3405

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Type II - Automatic - Powered stock feed

Class A - Non-adjustable cutting angle
(90 degrees from workpiece axis)

Size 1 - 12 x 12-inch workpiece capacity
2 - 12 x 16-inch workpiece capacity
3 - 12 x 20-inch workpiece capacity
4 - 14 x 17-inch workpiece capacity
5 - 15 x 20-inch workpiece capacity

Class B - Adjustable cutting angle
(90 degrees to 45 degrees from workpiece axis)

Size 2 - 12 x 16-inch workpiece capacity
4 - 14 x 17-inch workpiece capacity

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards
(29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/NFPA 79 - Metalworking Machine Tools and Plastics Machinery, Electrical Standards for (DoD adopted).

ANSI B11.10 - Safety Requirements for the Construction, Care, and Use of Metal Sawing Machines (DoD adopted).

(Application for copies should be addressed to American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for (DoD adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.2 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of performance production sawing operations in accordance with the requirements specified herein. Type I machines shall be designed for cutting bar stock where the workpiece is manually moved to the desired length for each cut. Type II machines shall be designed for repetitive, automatic indexing cutoff operation. Class A machines shall have a fixed cutoff at 90 degrees to workpiece axis. Class B machines shall be capable of cutoff at 90 degrees to 45 degrees to workpiece axis. Class B

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machines shall be designed to minimize space requirements and interference with adjacent equipment by setting the saw blade on an angle to the workpiece, rather than setting the workpiece on an angle to the machine, when performing miter cuts on long bars.

3.2.1 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.2 Energy efficiency. The machine and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.3 Controls. All operating controls shall be located convenient to the operator, and identified to facilitate selection and use.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the saw that present safety hazards. The safety devices shall not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specially required herein, shall comply with all of the requirement of OSHA 2206 that are applicable to the machine itself. In addition, the machine shall comply with all requirements of ANSI B11.10 that are designated therein as the responsibility of the machine manufacturer. In the event of conflict between the requirements of OSHA and the ANSI Standards, the requirements of OSHA shall apply. Additiona safety and health requirements shall be as specified (see 6.2.1).

3.2.5 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts (except sealed-for-life-bearings) that require continuous lubrication. Recirculating systems, when provided, shall include filters which are cleanable or replaceable. All oil holes, grease fittings, and filler caps shall be easily accessible.

3.2.6 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances which will permit installation of replacement parts without modification of part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

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3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All metal gears shall be machined and heat treated by a process that will impart the hardness, surface finish and toughness that will enable the gear train to transmit full-rated torque of the drive motor without gear damage, failure or premature deterioration and wear. All gears shall be manufactured to definite established standards to provide for interchangeability and parts replacement.

3.4 Components. The machine shall consist of a base, frame, band drive system, band wheels, band guides and guide arms, vise, hydraulic system, coolant system and such other components as are necessary to accomplish the performance requirements described herein.

3.4.1 Base.

3.4.1.1 Type I or II, class A. The base of type I or type II, class A machines shall be of heavy duty construction and shall support the vise,

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chip collector, coolant reservoir, coolant pump, hydraulic system, sawing frame and workpiece. Hinged access doors or removable panels shall be provided to facilitate servicing and adjusting internal components. Means shall be provided for leveling and securely anchoring the machine.

3.4.1.2 Type I or II, class B. The base of the type I or II, class B machine shall conform to all requirements for class A machines. In addition, either the sawing frame or the base upon which the saw frame is mounted shall swivel from 90 degrees to 45 degrees in relation to the work axis. Swiveling shall be accomplished by power. A graduated scale shall be provided for setting the angle of cut. Means shall be provided to securely lock the angle of cut. Means shall be provided to securely lock the swivel unit in set position for sawing.

3.4.2 Sawing frame. The frame unit shall contain the band guides and guide arms, automatic band tension device and saw band. A protective cover shall enclose the band wheels and saw blade so that a minimum of blade length is exposed for sawing. The cover, or guard, shall have hinged doors or other suitable means for providing ready access to the blade and wheels. A hydraulic system shall furnish power for raising and lowering the sawing frame and maintaining constant feed force. A pressure gauge or other device shall be provided to indicate the down feed pressure of the sawing frame.

3.4.3 Band drive system. The drive system shall provide infinitely variable band speeds. Gears or gear shafts utilized shall be supported by ball or roller bearings in completely enclosed housings, and shall have positive means of lubrication. A band speed indicator shall be provided to indicate speed of the band in feet per minute.

3.4.4 Band feed systems. The machine shall have a band feed system capable of providing infinitely variable feed forces. Operator's control devices shall be provided for selecting down feed, neutral, and retract functions over a range of 0 to not less than 300 pounds.

3.4.5 Band wheels. The band wheels shall be the flanged type. The idler wheel shall have a metal face or replaceable resilient tire and shall be adjustable to facilitate changing saw bands and to accommodate varying saw band lengths. The band wheels shall be protected from chips by a suitable band cleaning brush which is installed to clean chips from the saw band. The wheels shall be supported by ball or roller bearings which are either continuously lubricated or the sealed-for-life-type.

3.4.6 Band tensioning devices. An adjustable tensioning mechanism shall be installed on the idler wheel, capable of maintaining a band tension of up to 30,000 PSI either mechanically or hydraulically while the machine is in operation. Band tension may be applied automatically, or may be manually adjusted by turning a hand wheel to a marked setting. Automatically applied band tensioning mechanisms shall maintain a preset force while cutting, and

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relax band tension when stopped, without dislocating the blade on the wheels.

3.4.7 Band guides and guide arms. Band guides shall consist of solid carbide or carbide faced inserts, hardened and ground side rollers on each side of the saw band, or a combination of these, and a carbide back-up block on roller mounted or ball bearings. Two guide arms shall be attached to the sawing head for supporting the band guides which, in turn, provide the rigidity and stability required for the machine to perform production cutoff operations. One or both of the arms shall be adjustable for cutting various workpiece widths within the range of the machine. The adjustable arm(s) shall be easily positioned and shall be provided with locking mechanisms.

3.4.8 Worktable and vise. The worktable and vise shall be capable of supporting workpiece(s) within the rated capacity of the machine. The vise shall be power operated and shall have provision for adjusting the clamping pressure on the workpiece.

3.4.9 Automatic bar feed unit. Type II saws shall have an automatic stock feeding device, conforming to this paragraph, fitted to the saw. The automatic feeding device shall have a direct reading digital length readout calibrated in either inches or millimeters, or in combination. The automatic feeding device shall include a manual feature to make a single cut and a feature to accommodate multiple cuts over an incremental range of not less than 0-12 inches per single index. The feeding device shall also incorporate a multiple index feature to produce workpieces longer than 12 inches and up to 72 inches in length. The ways of feeding devices shall be either covered, enclosed, or have wipers and be lubricated by either a manual or automatic lubricating system. Automatic feeding devices shall include the number of power operated vises necessary to feed stock material (forward and reverse) through the automatic cycles to preclude manual push-up of the stock. When specified (see 6.2.1), the bar feed unit shall be equipped with a cycle counter which allows the operator to preset at the number of pieces to be cut before automatic machine shut down.

3.4.10 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant. Coolant system capacity shall be the manufacturer's standard, including the reservoir.

3.4.11 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining

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fluid level. The hydraulic system capacity shall be the manufacturer's standard, including the reservoir.

3.4.12 Electrical equipment. Unless otherwise specified (see 6.2.1), the electrical system shall conform to this paragraph. The electrical system shall conform to ANSI/NFPA No. 79 Electrical Standards for Metalworking Machine Tools. An identified terminal shall be provided for connection of the size grounding conductor specified in table 240-A of the NFPA Standard. The saw shall accept power from a 230/460-volt, 3-phase, 60-hertz AC system. Unless otherwise specified (see 6.2.1), the saw shall be initially wired for operation on 230 volts.

3.4.12.1 Motors. Motors shall have ball or roller bearings. Band drive motor horsepower shall meet or exceed the requirements of table I or table II. Unless otherwise specified (see 6.2.1), motors shall be single speed and meet or exceed the requirements of a drip proof enclosure.

3.4.12.2 Solid-state components. Solid-state design shall be used throughout for electronic components. The use of selenium or other similar ageing devices is permitted only for voltage surge protection to other solid-state components.

3.4.12.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 115 volts.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and characteristics of the machine shall be not less than those specified in table I or table II for the machine size specified.

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TABLE I. Sizes and capacities.

Characteristics	Manual Stock Feed					
	Type I - Class A					Type I - Class B
	Size 1	Size 2	Size 3	Size 4	Size 5	Size 2
Workpiece capacity, in:						
Rectangular	12 x 12	12 x 16	12 x 20	14 x 17	15 x 20	12 x 16
Round	12-3/4	16	16	14	15	16
Rectangular at 45°:	—	—	—	—	—	12 x 6-1/2
Round at 45°:	—	—	—	—	—	10
Band data:						
Width & thickness:	1-1/4 x .042	1-1/4 x .042	1-1/2 x .050	1-1/4 x .042	1-1/4 x .042	1-1/4 x .042
Optional:	1 x .035	1 x .035	—	—	1-1/2 x .050	1 x .035
Band drive motor, HP:	5	5	10	5	7-1/2	5
Optional HP:	—	—	—	—	10	—
Band feed force, infinitely variable:						
Pounds:	0 - 300	0 - 400	0 - 600	0 - 300	0 - 400	0 - 400
Optional:	—	—	—	—	0 - 600	—
Band speed range, infinitely variable, FPM:	60 - 350	60 - 350	60 - 350	60 - 450	55 - 400	60 - 350

NOTE: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine is offered, it shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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TABLE II. Sizes and capacities.

Characteristics	Automatic - Powered Stock Feed						
	Type II - Class A					Type II - Class B	
	Size 1	Size 2	Size 3	Size 4	Size 5	Size 2	Size 4
Workpiece capacity, in:							
Rectangular:	12 x 12	12 x 16	12 x 20	14 x 17	15 x 20	12 x 16	14 x 17
Round:	12-3/4	16	16	14	15	16	14
Rectangular at 45°:	—	—	—	—	—	12 x 6-1/2	9 x 11-3/4
Round at 45°:	—	—	—	—	—	10	9
Band data:							
Width & thickness:	1-1/4 x .042	1-1/4 x .042	1-1/2 x .050	1-1/4 x .042	1-1/4 x .042	1-1/4 x .042	1 x .035
Optional:	1 x .035	1 x .035	—	—	1-1/2 x .050	1 x .035	—
Band drive motor, HP:	5	5	10	5	7-1/2	5	5
Optional HP:	—	—	—	—	10	—	—
Band feed force, infinitely variable:							
Pounds:	0 - 300	0 - 400	0 - 600	0 - 300	0 - 400	0 - 400	0 - 400
Optional:	—	—	—	—	0 - 600	—	—
Band speed range, infinitely variable, FPM:	60 - 350	60 - 350	60 - 350	60 - 450	55 - 400	60 - 350	50 - 450

NOTE: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine is offered, it shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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3.6 Performance.

3.6.1 Metal removal rate and accuracy. The machine shall be capable of sawing SAE 1020 steel at a rate of 15 square inches per minute when tested in accordance with 4.5.2.1. Saw cuts produced at these rates shall be perpendicular to the surface resting on the worktable within ±0.002 inch per inch of thickness and shall be flat within ±0.002 inch per inch (concave or convex).

3.6.2 Automatic bar feed accuracy. Type II machines shall be capable of cutting the disks in 4.5.2.2 within ±0.005 inch of preset length. The thickness of the 12 disks shall vary not more than 0.010 inch.

3.6.3 Angular cut accuracy, class B machines. Class B machines shall be capable of cutting angles to within 1/2 degree of the selected angle to which the machine is set.

3.7 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. Hand tools. One set of hand tools normally furnished by the supplier.

b. Work indicator dial. A dial or chart indicating recommended cutting speeds, feed rates and types of band for general sawing operations.

c. Saw band. One high speed steel saw band of the type used in performance testing the machine.

d. Chip collection pan. A removable chip collection pan or drawer. on machines not equipped with chip conveyors.

3.8 Optional requirements. Optional equipment, accessories, and operating features required in addition to or in lieu of those required herein, shall be as specified (see 6.2.1).

3.9 Repair parts. Such repair parts as are specified (see 6.2.1) shall be furnished.

3.10 Marking of instruction plates, indicating plates, and nameplates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.10.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

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Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract.

3.12 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirement specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When first article approval is required under 3.1, first article inspection shall be performed. Unless otherwise specified (see 6.2.1), first article inspection shall be comprised of the examination in 4.4 and all tests in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the first article.

4.3 Quality conformance inspection. Each machine shall be subjected to quality conformance inspection prior to being offered for acceptance.

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Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5.1 through 4.5.2.3 and the inspection in 4.6. Failure to pass the examination or any test shall be cause for rejection.

4.4 Examination. The machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.5 and 3.7 through 3.12.

4.5 Tests.

4.5.1 Operational test. The machine shall be operated at no load for not less than 15 minutes. Proper operation of all controls, motors, adjusting mechanisms and accessories shall be verified during the trial period.

4.5.2 Performance test. The machine shall be subjected to the following tests to determine compliance with the performance and accuracy requirements of 3.6. For the purpose of these tests, sawing time shall include time consumed during stalls, but shall exclude time required for band changes. A high speed bi-metal saw band shall be used for all tests.

4.5.2.1 Bar stock sawing test. Each machine shall make twelve cuts through 5-inch diameter, SAE 1020 steel bar stock. The disk produced by the cuts shall be not less than 3/8-inch thick. The cutting speed rate and feed rate shall be at the option of the manufacturer. The time consumed while making the twelve cuts shall be determined and the sawing rate, flatness and squareness of the cut shall conform to 3.6.1.

4.5.2.2 Automatic bar feed accuracy test. Each machine provided with an automatic bar feed unit shall perform the test in 4.5.2.1 utilizing the automatic bar feed to make twelve cuts. Each disk cut shall meet the requirements of 3.6.2. On machines equipped with a cut counter, the counter shall be set for 12 cuts and the machine shall automatically stop after 12 disks have been cut.

4.5.2.3 Angular cut accuracy test (class B machines). Using the same type workpiece described in 4.5.2.1, the machine base or sawing frame, shall be swiveled and a cut shall be made at 15°, 30°, and 45°. Each angular cut shall meet the requirements of 3.6.3.

4.6 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging, packing, and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM-D-3951. When specified, preservation, packing, and marking shall be in accordance with specification MIL-M-18058. Level of preservation and packing shall be as specified (see 6.2.1).

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6. NOTES

6.1 Intended use. Machines covered by this specification are intended for use in heavy duty, production, high speed cutoff operations. Class A machines provide straight cutoff at 90 degrees to workpiece axis. Class B machines provide for straight (90 degree) cutoff or angular (0 - 45 degree) cutoff to the workpiece axis.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Type, class, and size band saw required (see 1.2).

c. First article approval, if required (see 3.1).

d. Specify additional safety and health requirements, (see 3.2.4 and 6.4).

e. Painting, if different (see 3.3.5).

f. Bar feed cycle counter for type II machines if repaired (see 3.4.9).

g. Electrical system and voltage for initial wiring, if different (see 3.4.12).

h. Motor size and type (see 3.4.12.1 and table I or II).

i. Sizes and capacities, if different (see 3.5).

j. Standard equipment, if different (see 3.7).

k. Optional requirements, if required (see 3.8).

l. Repair parts, if required, as specified (see 3.9).

m. Lubrication chart, if different (see 3.10.1).

n. Nameplate, if different (see 3.10.2).

o. First article inspection, if different (see 4.2).

p. Quality conformance inspection, if different (see 4.3).

q. Required level of preservation and packing (see 5.1).

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6.2.2 Contract data requirements. All data requirements, such as parts lists, wiring diagrams, operating manuals, maintenance manuals, alignment and test results, shall be requested on a DD Form 1423, Contract Data Requirements List, and shall be furnished in accordance with each specified DD Form 1664, Data Item Description, as shown on the DD Form 1423.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all points-of-operation guarding normally furnished as standard on the manufacturer's commercial machines supplied to the commercial market. If specified point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required as, in most cases, the guard configuration is dependent upon the size and configuration of the workpieces. The above, and any other additional safety and health requirements, should be specified in detail under 6.2.1.d.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activities:
Army - AL
Navy - SH
Air Force 84

User Activities:
Army - SM
Navy - MC

Preparing Activity:
DLA-IP

Project Number:
3405-0073

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-24036D	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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PREVIOUS EDITION IS OBSOLETE.

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